FUSE-QA: Improving Long-Tail Question Answering with Hybrid Retrieval and Distilled Evidence Filtering

Anonymous EMNLP submission

Abstract

Long-tail question answering remains challenging for retrieval-augmented generation because standard pipelines often fail in two ways: they miss relevant evidence for rare entities during retrieval and pass noisy context to the generator even when relevant passages are found. We present a retrieval-to-generation framework for long-tail QA that improves both evidence coverage and evidence precision. First, we propose FUSE-QA, a hybrid multi-stage retriever that combines BM25 and DPR with reciprocal rank fusion and cross-encoder reranking, improving evidence retrieval without retriever fine-tuning or query expansion. Second, we introduce DistillSRE, a lightweight sentence-level relevance estimator distilled from a stronger instructor model, which filters retrieved passages to retain answer-bearing sentences before generation. Experiments on POPQA and ENTITYQUESTIONS show that our framework consistently improves both retrieval recall and end-to-end QA accuracy on long-tail questions. On long-tail POPQA, LLaMA-3 8B with FUSE-QA + DistillSRE achieves 47.3% exact match, outperforming the recent RPDR (41.4%) baseline by 5.9 points. These results show that stronger retrieval and cleaner evidence can matter more than model scale for rare-entity question answering.

1 Introduction

Large language models (LLMs) have achieved impressive performance across a wide range of question answering tasks, yet remain unreliable on questions involving *long-tail knowledge* (Dai et al., 2023a; Sun et al., 2024; Li et al., 2024). Many real-world queries concern rare entities or infrequent facts that are weakly represented in pretraining corpora, causing purely parametric models to produce unsupported or hallucinated answers (Kandpal et al., 2023a; Dai et al., 2023b). This limitation has motivated retrieval-augmented generation (RAG), in which external documents are retrieved and provided as evidence during answer generation (Lewis et al., 2020; Jiang et al., 2025).

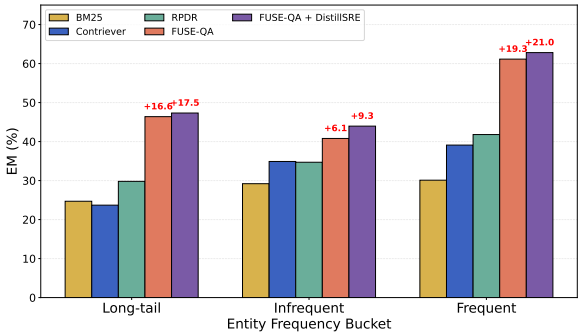


Figure 1: Long-tail QA is challenging for standard RAG pipelines because sparse and dense retrieval capture complementary evidence, while concatenated passages introduce distracting context. FUSE-QA fuses sparse and dense retrieval, and DistillSRE filters sentence-level evidence. On PopQA with LLaMA-3 8B, the approach improves exact match across entity-frequency tiers, with additional gains over RPDR.

However, long-tail QA is not solved simply by adding retrieval. As illustrated in Figure 1, existing RAG pipelines often struggle because relevant evidence for long-tail questions may be missed during retrieval, and even when relevant passages are retrieved, they frequently contain substantial distracting context for generation. More broadly, long-tail QA remains difficult for three main reasons. First, LLMs have limited parametric coverage of rare entities and infrequent facts, making external evidence essential for reliable answering (Kandpal et al., 2023a; Mallen et al., 2023b). Second, retrieval itself remains unreliable for rare-entity questions: sparse retrievers such as BM25 (Robertson and Zaragoza, 2009a) depend on lexical overlap, whereas dense retrievers such as Dense Passage Retrieval (DPR) (Karpukhin et al., 2020a) do not generalize well to rare or out-of-distribution entities (Sciavolino et al., 2021a; Graciotti et al., 2025; Qi et al., 2025). Third, retrieved passages are often too broad for effective generation, containing sub-

stantial irrelevant context even though only a few sentences provide answer-bearing evidence (Wang et al., 2025; Zhang et al., 2025). Taken together, these challenges suggest that long-tail QA requires both stronger evidence retrieval and more precise evidence selection for grounded generation.

Earlier work has focused primarily on improving the retrieval of rare entities. For example, RPDR (Zhang et al., 2025) improves dense retrieval through synthetic data augmentation and rare-entity-aware retriever training. However, long-tail knowledge remains difficult in retrieval-augmented QA: rare and emerging questions continue to pose challenges even when external retrieval is available (Zhang et al., 2024). Entity-centric approaches have also improved evidence acquisition by retrieving entity-linked knowledge sources (Shavarani and Sarkar, 2024). While effective, these methods remain largely retrieval-centric: they focus on which documents or passages to retrieve, but pay less attention to *how retrieved evidence should be refined before being passed to the generator*.

In this work, we argue that long-tail QA requires a complete *retrieval-to-generation* solution rather than a retriever-only fix. To this end, we propose a two-stage framework that improves both *evidence coverage* and *evidence precision*. First, to address the retrieval bottleneck for rare entities, we introduce FUSE-QA, a hybrid retrieval pipeline that combines BM25 and DPR through Reciprocal Rank Fusion (RRF) and cross-encoder reranking. This setup enhances recall without the need for costly retriever fine-tuning. Second, to prevent noisy context from distracting the reader, we propose DistillSRE, a lightweight sentence-level relevance estimator. By applying knowledge distillation, DistillSRE filters retrieved documents to retain only answer-bearing evidence before generation. We evaluate our approach on PopQA (Mallen et al., 2023a) and EntityQuestions (Sciavolino et al., 2021a). As shown in Figure 1, our method yields strong gains across entity-frequency tiers, with especially large improvements on Long-tail and Infrequent questions. Notably, Llama-3 8B with **FUSE-QA + DistillSRE** achieves **47.3%** EM on long-tail PopQA, surpassing GPT-3.5 + RPDR (41.4%) and showing that improving retrieval quality and evidence precision can matter more than raw model scale for rare-entity question answering. Our contributions are summarized as follows:

- We reframe long-tail QA as a joint optimization of *retrieval coverage* and *generation precision*, identifying that sparse-dense hybrid retrieval must be paired with granular evidence filtering to handle rare entities effectively.
- We introduce **FUSE-QA**, a multi-stage framework that yields superior recall on rare-entity questions by fusing sparse and dense signals followed by reranking, notably achieving these gains without task-specific retriever fine-tuning.
- We propose DistillSRE, a sentence-level relevance estimator that employs knowledge distillation to compress retrieved context into high-density, answer-bearing evidence, significantly reducing generation-time noise.
- Our framework consistently outperforms strong baselines on PopQA and EntityQuestions. Most notably, our Llama-3 8B-based pipeline surpasses GPT-3.5-scale models (e.g., +5.9% EM on long-tail PopQA), demonstrating that high-precision retrieval can compensate for model scale in the long tail.

2 Related Work

Long-tail knowledge in LLMs. Large language models exhibit strong performance on frequently encountered facts but remain unreliable on questions involving rare entities and infrequent knowledge (Kandpal et al., 2023b; Mallen et al., 2023c). Kandpal et al. (2023b) show that an entity's pretraining frequency strongly predicts whether an LLM can answer questions about it, highlighting the long-tail gap in parametric memory. Retrieval-augmented generation (RAG) has emerged as a practical remedy by grounding generation in external evidence (Lewis et al., 2020; Asai et al., 2023), yet standard RAG pipelines still struggle on long-tail queries because the retrieval itself may fail for rare entities (Sciavolino et al., 2021b; Leszczynski et al., 2022; Tran et al., 2025). Our work builds on these findings and targets the full retrieval-to-generation pathway for long-tail QA, rather than addressing either component in isolation.

Dense and hybrid retrieval. Dense retrieval models encode queries and passages into shared vector spaces for semantic matching (Karpukhin et al., 2020b; Izacard et al., 2021; Xiong et al., 2020). More recent embedding models such as BGE (Xiao et al., 2023), NV-Embed (Lee et al., 2024), and Gemma embeddings (Team et al., 2024) have achieved strong results on general retrieval benchmarks, yet they still generalize poorly to rare

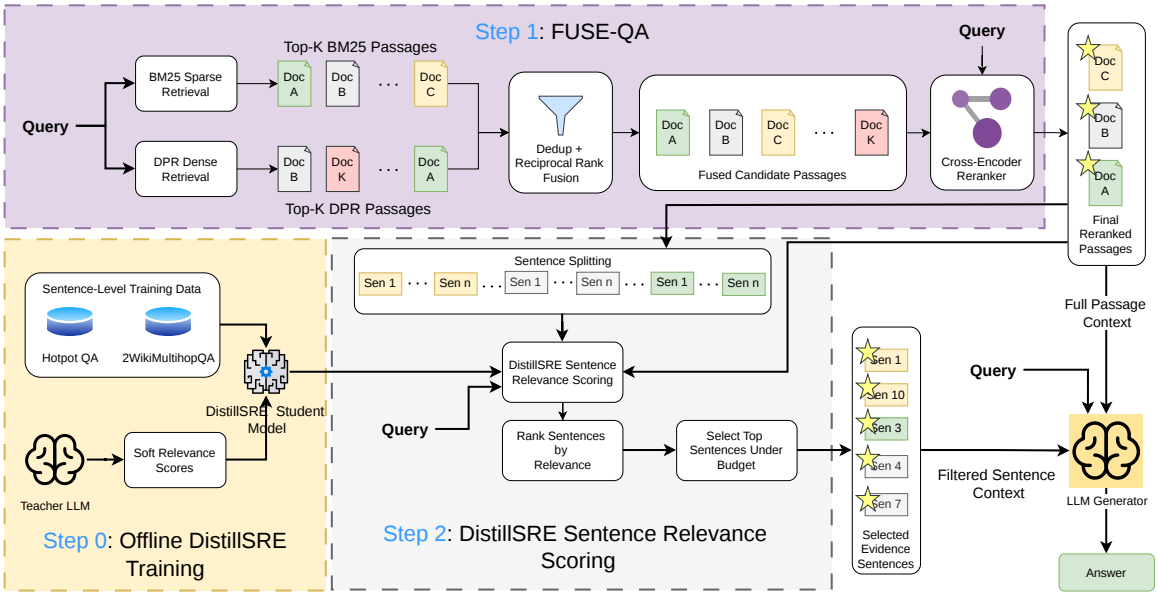


Figure 2: Overview of our retrieval-to-generation framework for long-tail question answering. *Step 0: Offline DistillSRE Training* learns a sentence relevance estimator from sentence-level QA supervision augmented with teacher-provided soft relevance scores. *Step 1: FUSE-QA* performs hybrid retrieval by combining BM25 and DPR, merges candidates using deduplication and reciprocal rank fusion, and applies cross-encoder reranking to produce high-quality evidence passages. *Step 2: DistillSRE Filtering* splits reranked passages into sentences, scores each sentence with $p_i = \mathrm{SRE}(q, D, s_i)$, and selects the top evidence under a fixed budget. The generator then answers using either the full passages or the filtered sentences, where filtering increases evidence density and reduces distracting context.

or out-of-distribution entities (Sciavolino et al., 2021b). Zhang et al. (2025) address this gap with RPDR, a data augmentation framework that uses round-trip prediction to select high-quality synthetic training examples for improving dense retrievers on long-tail questions. While RPDR and related approaches focus on training better dense retrievers, ours takes a complementary direction: we combine an off-the-shelf dense retriever with sparse retrieval via reciprocal rank fusion (Cormack et al., 2009b) and cross-encoder reranking (Xiao et al., 2023), achieving competitive or superior retrieval recall *without* task-specific retriever fine-tuning. Hybrid sparse–dense retrieval has been explored in broader information retrieval settings (Lin et al., 2021a; Robertson and Zaragoza, 2009b), but its effectiveness specifically for long-tail entity-centric QA has received limited attention.

Evidence filtering and context compression for RAG. Even when retrieval succeeds, concatenated passages often contain distracting context that degrades generation quality (Shi et al., 2023; Amiraz et al., 2025). Prior work addresses this through various forms of context compression. RECOMP (Xu et al., 2023) proposes both extractive and abstractive compressors that select or summarize retrieved sentences before generation. EXIT (Hwang et al., 2025) introduces a context-aware extractive framework that classifies sentences while preserving contextual dependencies. Self-RAG (Asai et al., 2023) takes a different approach by training the generator itself to decide when to retrieve and to critique the relevance of retrieved passages via special reflection tokens. Other methods compress retrieved context using evidentiality-guided filtering (Jeong et al., 2025) or familiarity-aware evidence compression (Jung et al., 2025). Our DistillSRE differs from these approaches in two aspects: it operates at the sentence level with a lightweight classifier distilled from a stronger teacher model, and it is specifically designed for entity-centric long-tail QA where the answer is typically supported by one or a few sentences within longer passages.

Knowledge distillation for NLP. Knowledge distillation (Hinton et al., 2015) transfers knowledge from a larger teacher model to a smaller student, and has been widely adopted in NLP for model compression and task specialization. Hsieh et al. (2023) demonstrate that distilling step-by-step

reasoning from large models enables smaller models to outperform their teachers with less training data. In the retrieval setting, Izacard and Grave (2020) distill knowledge from reader models to retrievers for open-domain QA. Our use of distillation targets a different component of the RAG pipeline: we distill sentence-level relevance judgments from an instruction-tuned teacher into a lightweight estimator that filters evidence before generation, combining supervised labels with soft teacher scores to capture fine-grained distinctions between useful and distracting sentences.

3 Method

In this section, we describe the main components of our framework. As shown in Figure 2, our framework contains two main components: hybrid retrieval with reranking (§3.1) and sentence-level evidence filtering (§3.2).

3.1 Hybrid Sparse–Dense Retrieval

Sparse Retrieval. We first perform lexical retrieval using BM25 over a Wikipedia corpus to obtain an initial pool of candidate passages for a given question. Specifically, we retrieve the top K_{BM25} passages ranked by BM25 score, forming the lexical candidate set $R_{\text{lex}}(q)$. The BM25 index is built using Pyserini (Lin et al., 2021b) with parameter settings $k_1 = 0.9$ and $b = 0.4$.

Dense Retrieval. Alongside lexical retrieval, we use DPR (Karpukhin et al., 2020c). We build a DPR-based index over Wikipedia by encoding each article from its title and body text, truncated to the first 100 tokens. We use the pooled output as the passage representation, ℓ_2-normalize all passage embeddings, and index them in FAISS (Johnson et al., 2019) for inner-product search. Given a question q, we encode it with a pretrained DPR question encoder and retrieve the top K_{DPR} passages from the index, forming the dense candidate set $R_{\text{dense}}(q)$. We construct the index offline and reuse it at inference time; Appendix B.3 gives the implementation details.

Reciprocal Rank Fusion. To combine lexical and dense retrieval, we apply reciprocal rank fusion (RRF) (Cormack et al., 2009b) to their merged candidate set:

$$R_{\text{union}}(q) = R_{\text{lex}}(q) \cup R_{\text{dense}}(q). \tag{1}$$

We deduplicate passages retrieved by both methods using normalized titles and passages before fusion. Each passage $p \in R_{\text{union}}(q)$ receives the score:

$$s_{\text{RRF}}(p) = \sum_{r \in \{\text{BM25,DPR}\}} \frac{1}{\gamma + \text{rank}_r(p)}, \tag{2}$$

where $\text{rank}_r(p)$ is the rank of p under retriever r, and γ is a smoothing constant. Following prior work (Cormack et al., 2009a), we set $\gamma = 60$.

Hybrid retrieval combines sparse lexical retrieval and dense semantic retrieval in a single pipeline. In our framework, BM25 captures exact lexical matches while DPR captures semantic similarity. RRF then fuses the rankings from both retrievers, improving robustness. This is particularly beneficial for long-tail questions, where rare entities may be easier to recover through lexical matching than through dense retrieval alone.

Cross-Encoder Reranking. Because the fused candidate set is optimized for recall, it may still contain weakly related passages. To improve precision, we rerank the top $\tilde{K}$ fused candidates using a cross-encoder reranker. We use the BGE reranker (Xiao et al., 2023), a RoBERTa-based sequence classification model that jointly encodes each question–passage pair (q, p) and outputs a relevance score $s_{\text{CE}}(q, p)$. We then retain the top $k = 30$ passages as the final evidence set:

$$E(q) = \{p_1, \ldots, p_k\}. \tag{3}$$

The resulting evidence set is passed to the sentence filtering stage and subsequently used for answer generation.

3.2 Sentence-Level Evidence Filtering

Even when relevant passages are retrieved, they often contain substantial irrelevant or distracting content. This issue is particularly pronounced for long-tail questions, where retrieved passages may mention the queried entity many times but still fail to contain the specific fact needed to answer the question. For example, a biography page may repeatedly mention the entity while only one sentence actually states the answer (see Figure 3). To address this problem, we introduce DistillSRE, a lightweight distilled sentence relevance estimator that identifies answer-bearing sentences given a question and its retrieved context.

Formally, DistillSRE is a classifier $g_\theta(q, D, s)$ that takes a question q, retrieved context D, and candidate sentence s as input, and outputs the probability that s is useful for answering q in the context

of D. We cast this as a binary classification problem with label $y \in \{0, 1\}$, where $y = 1$ indicates useful evidence. At inference time, DistillSRE is applied to sentences extracted from the reranked passages produced by FUSE-QA to remove noisy context before generation.

Training Data and Distillation. We train DistillSRE using QA datasets with sentence-level supporting fact annotations, specifically HotpotQA (Yang et al., 2018) and 2WikiMultiHopQA (Ho et al., 2020). For each question, we expand the associated Wikipedia context into candidate sentences and derive labels from the annotated supporting facts. Supporting-fact sentences are treated as positive evidence, while non-supporting sentences are used as negatives. To provide richer supervision beyond binary labels, we further distill knowledge from a stronger instruction-tuned teacher model (Hinton et al., 2015; Hsieh et al., 2023). Given (q, D, s), the teacher predicts a probability distribution over the tokens "Yes" and "No", and we denote the teacher's soft relevance score as

$$\pi_{\text{yes}} = P_{\text{teacher}}(\text{“Yes”} \mid q, D, s). \tag{4}$$

These soft scores provide graded supervision, allowing the student to distinguish irrelevant sentences from harder negatives that are topically related but not answer-bearing. Additional details on training instance construction and teacher-guided supervision are provided in Appendix A.

Training Objective. Let $p_i = P_\theta(y = 1 \mid q_i, D_i, s_i)$ denote the student model's predicted probability that sentence s_i is useful for answering question q_i given context D_i, and let $\hat{\pi}_i = P_\theta(\cdot \mid q_i, D_i, s_i)$ denote the student's predictive distribution. We train DistillSRE with a combination of supervised and distillation-based objectives:

$$\mathcal{L}_{\text{SRE}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \Big[(1 - \lambda)\, \mathcal{L}_{\text{CE}}(y_i, p_i) + \lambda\, \mathcal{L}_{\text{KD}}(\pi_i, \hat{\pi}_i) \Big]. \tag{5}$$

where $\mathcal{L}_{\text{CE}}$ is binary cross-entropy with ground-truth label y_i, and $\mathcal{L}_{\text{KD}}$ encourages the student predictive distribution $\hat{\pi}_i$ to match the teacher's soft relevance distribution π_i. We set $\lambda = 0.5$ in all experiments.

Inference and Context Selection. At inference time, DistillSRE is applied to candidate sentences extracted from the reranked passages returned by FUSE-QA. For each candidate sentence s, the model computes a relevance score $P_\theta(y = 1 \mid q, D, s)$, and all candidate sentences are ranked accordingly. We then retain the highest-scoring sentences under a fixed budget and concatenate them as the final evidence context for answer generation. This filtering step removes distracting content while preserving the most useful evidence, yielding shorter and more focused inputs for the generator. Full implementation details, including model architecture, training setup, and inference budget, are provided in Appendix A.

4 Experiments

We evaluate our framework on both retrieval quality and end-to-end question answering over two long-tail QA benchmarks. We first describe the experimental setup, then report retrieval and answer-generation results across entity-frequency tiers. Overall, the experiments show that FUSE-QA improves evidence coverage and ranking quality, while DistillSRE further improves the precision of the evidence presented to the generator.

4.1 Experimental Setup

Datasets. We evaluate on PopQA (Mallen et al., 2023a) and EntityQuestions (Sciavolino et al., 2021a), two open-domain QA benchmarks grounded in Wikipedia. Following the evaluation protocol used in RPDR, we group questions into three entity-frequency tiers based on subject popularity: **Long-tail** ($s_{pop} < 100$), **Infrequent** ($100 \leq s_{pop} < 10{,}000$), and **Frequent** ($s_{pop} \geq 10{,}000$). PopQA provides subject popularity annotations directly. EntityQuestions does not, so we estimate subject popularity from Wikipedia pageview statistics and apply the same thresholds. Additional dataset details and tier construction are provided in Appendix B.1. The resulting evaluation distribution across tiers is shown in Table 2.

Retrieval, filtering, and generators. FUSE-QA uses a BM25 index over Wikipedia (≈21M passages) built with Pyserini (Lin et al., 2021b), together with a DPR dense index over 100-token Wikipedia passages encoded using a pre-trained DPR model without task-specific fine-tuning (Karpukhin et al., 2020a). The two ranked lists are merged with reciprocal rank fusion and reranked using the off-the-shelf cross-encoder `BAAI/bge-reranker-base` (Xiao et al., 2023). In

Generator	Retriever	PopQA (EM, %)			EntityQuestions (EM, %)		
		Long-tail	Infrequent	Frequent	Long-tail	Infrequent	Frequent
GPT-5-NANO	Parametric (no retr.)	24.73	10.41	29.31	16.57	17.85	19.11
	FUSE-QA (ours)	37.67	21.90	34.56	37.73	36.64	26.64
	FUSE-QA + DistillSRE (ours)	37.46	20.97	33.29	37.12	35.91	28.50
QWEN2.5-7B	Parametric (no retr.)	17.44	9.25	36.62	8.31	8.12	12.77
	FUSE-QA (ours)	43.82	40.45	57.03	40.53	44.38	38.06
	FUSE-QA + DistillSRE (ours)	44.53	40.83	56.81	**41.38**	**45.45**	39.82
LLAMA-3-8B	Parametric (no retr.)	11.90	21.60	29.90	10.30	14.50	21.70
	BM25	24.70	29.20	30.10	27.80	29.60	32.80
	Contriever (dense)	23.70	34.90	39.10	21.80	25.40	30.20
	RPDR	29.80	34.70	41.80	30.10	31.60	33.90
	FUSE-QA (ours)	46.39	40.80	61.14	37.66	43.07	38.89
	FUSE-QA + DistillSRE (ours)	**47.32**	**43.96**	62.81	39.15	42.18	**40.29**
QWEN3-8B	Parametric (no retr.)	16.15	7.39	33.85	10.15	11.80	19.47
	FUSE-QA (ours)	34.95	34.89	49.65	29.55	34.47	29.70
	FUSE-QA + DistillSRE (ours)	36.10	36.79	51.60	30.77	36.45	31.83
MIXTRAL-8X7B	Parametric (no retr.)	14.37	17.94	48.37	18.35	25.84	32.71
	FUSE-QA (ours)	33.81	35.79	51.56	39.81	44.08	36.55
	FUSE-QA + DistillSRE (ours)	34.10	35.76	50.14	39.85	43.75	35.98
LLAMA-3-70B	Parametric (no retr.)	24.02	30.55	**65.19**	20.90	29.69	36.86
	FUSE-QA (ours)	39.51	40.26	57.17	35.67	41.12	36.19
	FUSE-QA + DistillSRE (ours)	39.39	40.23	56.28	36.73	42.81	38.47
GPT-3.5	Parametric (no retr.)	22.70	36.40	50.30	20.60	29.90	36.80
	BM25	30.50	31.80	42.60	35.90	35.40	35.60
	Contriever (dense)	29.70	41.50	52.70	32.50	37.80	38.40
	RPDR	41.40	42.90	52.10	36.40	38.60	38.70

Table 1: **End-to-end exact match (EM, %) on POPQA and ENTITYQUESTIONS**, grouped by entity-frequency tier. Shaded entries are reproduced from (Zhang et al., 2025); GPT-3.5 is included only as a reference baseline from that work. **Best** results per column are shown in **bold**; second-best results are underlined. "Parametric" denotes answering without retrieval; FUSE-QA denotes our hybrid retrieval + reranking pipeline; DistillSRE applies sentence-level evidence filtering.

Tier	PopQA	EntityQuestions
Long-tail	1,399	2,209
Infrequent	10,050	13,198
Frequent	2,818	6,668
Total	14,267	22,075

Table 2: Distribution of evaluation questions across entity-frequency tiers.

our experiments, we retrieve the top 100 BM25 passages and top 20 DPR passages, retain up to 200 fused candidates, and rerank them to obtain the final passage set.

For sentence-level evidence selection, we apply DistillSRE to the reranked passages returned by FUSE-QA. DistillSRE scores candidate sentences for question relevance, and we keep the highest-ranked sentences under a fixed evidence budget. In our main experiments, we retain at most three sentences with a total budget of approximately 500 characters. For end-to-end QA, we evaluate GPT-5-NANO, QWEN2.5-7B, LLAMA-3-8B, QWEN3-8B, MIXTRAL-8X7B, and LLAMA-3-70B. Prompting details and evaluation settings are provided in Appendix B.2.

Metrics and baselines. For retrieval evaluation, we report recall at rank 10 and 20 (R@10, R@20). A retrieved passage is counted as correct if its text contains one of the gold answer strings. For end-to-end QA, we report Exact Match (EM) after standard answer normalization. We compare FUSE-QA against strong sparse and dense retrievers, including BM25, Contriever, BGE, NV-Embed, Gemma, RPDR, and RPDR-Random. For generation, we also report parametric answering without retrieval when available.

All generators are evaluated with the same answer-generation prompt template for each retrieval setting to ensure fair comparison across models. For retrieval-based generation, all methods are evaluated under the same evidence budget, and DistillSRE uses the same 500-character budget in all reported experiments. The prompt template is in Appendix B.2. Shaded RPDR results are copied

Method	PopQA						EntityQuestions					
	Long-tail		Infrequent		Frequent		Long-tail		Infrequent		Frequent	
	R@10	R@20	R@10	R@20	R@10	R@20	R@10	R@20	R@10	R@20	R@10	R@20
Baselines (reproduced from Zhang et al. (2025))												
BM25	62.6	66.8	46.2	52.8	53.8	61.1	48.3	63.5	41.9	66.7	55.0	69.2
Contriever	52.3	59.3	53.3	61.1	80.0	85.8	42.9	53.3	53.4	61.0	57.3	67.6
BGE	53.9	61.2	57.4	65.3	82.4	87.9	44.7	60.3	55.1	69.8	60.9	75.2
NV-Embed	55.0	62.3	63.5	70.6	89.9	90.6	48.4	61.6	57.1	70.7	71.5	84.3
Gemma	53.8	61.7	66.9	69.3	88.7	90.2	49.3	61.1	58.2	71.5	70.5	82.7
RPDR	74.5	75.7	76.8	78.2	81.3	87.1	54.6	68.1	57.4	71.2	62.7	70.0
RPDR-Random	66.8	70.0	70.0	74.5	82.7	87.3	51.9	67.7	52.4	69.5	55.9	70.8
Ablation: incremental components (ours)												
DPR only	58.4	63.9	55.6	62.2	77.0	81.4	39.8	47.9	46.0	53.1	46.5	53.3
Hybrid Retrieval	67.2	72.1	61.9	68.7	79.4	85.1	61.2	68.3	70.7	76.0	72.2	78.7
FUSE-QA (+Reranker)	**74.5**	**76.9**	**75.2**	**77.4**	**90.2**	**91.4**	**69.4**	**73.1**	**77.8**	**80.6**	**81.4**	**84.5**

Table 3: **Retrieval recall (R@10 / R@20) on POPQA and ENTITYQUESTIONS.** Shaded cells indicate results reproduced from Zhang et al. (2025). The ablation block isolates each component: DPR alone, hybrid retrieval (BM25 + DPR via RRF) without reranking, and full FUSE-QA with cross-encoder reranking.

directly from Zhang et al. (2025) rather than rerun. In particular, the GPT-3.5 rows in Table 1 come directly from RPDR and are included as reference baselines for comparison with prior work.

4.2 Retrieval Results

Table 3 reports retrieval recall on POPQA and ENTITYQUESTIONS. The results show a consistent pattern: hybrid retrieval substantially improves over DPR alone, especially on Long-tail and Infrequent questions. This confirms that sparse and dense retrieval provide complementary signals for rare-entity QA: BM25 helps recover passages through lexical overlap, while DPR helps when the question and evidence use different wording.

The ablation results further clarify the contribution of each stage. DPR alone provides a reasonable dense baseline, but recall drops markedly on harder frequency buckets. Adding BM25 and merging the two ranked lists with RRF yields a substantial improvement, confirming that hybrid retrieval increases evidence coverage for rare entities. Applying a cross-encoder reranker on top of the fused candidates yields the strongest results overall, showing that FUSE-QA benefits from both broader candidate coverage and more precise final ranking. The largest retrieval gains appear in the harder tiers. On both POPQA and ENTITYQUESTIONS, FUSE-QA achieves its strongest improvements on Long-tail and Infrequent questions, where retrieval is most difficult and where better evidence coverage matters most. These results indicate that the full retrieval pipeline improves both recall and ranking quality rather than relying on a single strong stage.

4.3 End-to-End QA Results

Table 1 reports end-to-end EM across generators and entity-frequency tiers. Overall, retrieval augmentation substantially improves over parametric answering alone, especially on Long-tail and Infrequent questions. This confirms that even strong modern generators still depend heavily on external evidence for rare-entity QA.

Across most generators, FUSE-QA improves answer accuracy relative to both retrieval-free baselines and prior retrieval baselines. The gains in retrieval quality therefore transfer directly into better final answers. In many cases, DistillSRE provides a further improvement by filtering noisy retrieved context and presenting the generator with a smaller, more focused evidence set.

The gains from DistillSRE are most visible for smaller and mid-sized models such as QWEN2.5-7B, LLAMA-3-8B, and QWEN3-8B. For these models, both retrieval quality and evidence precision strongly affect final answer accuracy. In several settings, the full FUSE-QA + DistillSRE pipeline enables these models to approach or surpass larger retrieval-based baselines on Long-tail questions. By contrast, the benefit of sentence filtering is less uniform for the strongest large models. For example, LLAMA-3-70B already performs strongly on **Frequent** questions even without DistillSRE, suggesting that powerful large models can often use longer retrieved context effectively when facts are easier to recover and verify.

Query: *In what city was John XIX born?*
Gold answer: Rome

Retrieved context before DistillSRE

Pope John XIX Pope John XIX (; died October 1032) was Pope from May 1024 to his death in 1032. Born Romanus in ***Rome*** *was the third son of Gregory I, Count of Tusculum and his wife Mary. While his brother Theophylactus held the papacy as Pope Benedict VIII, Romanus held the temporal power in the city as consul and senator. Upon the death of Benedict, Romanus, a layman, was elected to succeed him. He was immediately ordained in all the orders in succession, and consecrated bishop in order to enable him to ascend the papal chair . . .*

Selected evidence after DistillSRE

Born Romanus in ***Rome*** *was the third son of Gregory I, Count of Tusculum and his wife Mary.*

Score: 0.9546

Figure 3: Example showing how FUSE-QA and DistillSRE work together. The top-ranked FUSE-QA passage contains the correct answer but also includes noisy context, while the top-ranked DistillSRE sentence keeps only the part that directly answers the question.

4.4 Qualitative Example

Figure 3 shows a representative example. For the question *In what city was John XIX born?*, the top-ranked FUSE-QA passage already contains the correct answer, but it is embedded in a much longer biographical passage. DistillSRE assigns the highest score to the sentence that directly supports the gold answer and removes the surrounding details. This example illustrates the complementary roles of the two components: FUSE-QA retrieves the right passage, while DistillSRE refines it into a more useful evidence unit for generation. A more detailed worked example, including the top-ranked candidate sentences, is provided in Appendix B.4.

4.5 Discussion

Taken together, the results show that long-tail QA is a retrieval-to-generation problem rather than a retrieval-only problem. FUSE-QA mainly improves evidence coverage and ranking quality, while DistillSRE mainly improves evidence precision by removing distracting context. The benefit of sentence filtering is most consistent for smaller and mid-sized generators, and for harder Long-tail and Infrequent settings where noisy context is especially harmful.

4.6 Efficiency Study

We conduct a lightweight efficiency analysis on a stratified sample of PopQA ($n = 300$) using Llama-3.1-8B served by vLLM, comparing FUSE-QA and FUSE-QA+DistillSRE. For each example, we log final context length (characters, tokens, selected sentences), exact match (EM), and stage-level wall-clock times. DistillSRE substantially compresses the generator input: average context length drops from 752.5 to 555.6 characters (-26.2%), from 184.4 to 134.3 tokens (-27.2%), and selected sentences from 6.55 to 4.91 (-25.0%). Generation latency is nearly unchanged (0.0979s vs. 0.0957s), while DistillSRE introduces an additional filtering cost of 0.1128s, increasing total measured time from 0.0995s to 0.2103s. In this run, EM changes from 0.320 to 0.297. These timings are best-effort measurements with `time.perf_counter()`, and retrieval/fusion/reranking are precomputed in the dataset (thus their online timing is not fully represented).

5 Conclusion

We presented **FUSE-QA**, a retrieval-to-generation framework for long-tail question answering. Our results show that long-tail QA requires not only stronger evidence retrieval, but also more precise evidence selection for generation. FUSE-QA improves evidence coverage by combining BM25 and DPR with reciprocal rank fusion and cross-encoder reranking, while DistillSRE improves evidence precision by filtering noisy retrieved passages down to the most answer-bearing sentences.

Experiments on PopQA and EntityQuestions show that the full **FUSE-QA + DistillSRE** pipeline consistently improves both retrieval recall and end-to-end QA accuracy, with the largest gains on Long-tail and Infrequent questions. Notably, on Long-tail PopQA, our Llama-3-8B pipeline achieves 47.3% EM, outperforming GPT-3.5 + RPDR by 5.9 points, which highlights that better retrieval and cleaner evidence can matter more than model scale alone.

Future Work. Promising directions include adaptive retrieval that skips retrieval when parametric memory is sufficient, finer-grained evidence filtering at the answer-span level, and tighter integration between retrieval, filtering, and generation.

Limitation

Our study focuses on English, Wikipedia-grounded open-domain QA, evaluated on POPQA and ENTITYQUESTIONS. These benchmarks are well suited for studying long-tail entity-centric QA, but they do not cover broader settings such as multilingual QA, domain-specific corpora, or open-web retrieval. Additional validation is therefore needed to assess how well our findings generalize beyond Wikipedia-based long-tail QA.

Our framework also makes several deliberate design choices for simplicity and reproducibility. In particular, the dense retriever is built from an off-the-shelf DPR model without task-specific fine-tuning, and DistillSRE uses fixed-budget sentence filtering. These choices make the pipeline lightweight and easy to reproduce, though future work could explore more adaptive retrieval and filtering strategies. Finally, our evaluation is limited to short-answer factual QA, and we leave broader generation settings for future work.

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A Details of DistillSRE Training and Inference

Training Instance Construction. We construct DistillSRE training instances from QA datasets with sentence-level supporting fact annotations, specifically HOTPOTQA (Yang et al., 2018) and 2WIKIMULTIHOPQA (Ho et al., 2020). For each question, we expand the associated Wikipedia context into candidate sentences and derive labels from the supporting-fact annotations. Sentences annotated as supporting facts are treated as positive evidence; sentences from the same document that are not annotated as supporting facts are treated as hard negatives; and sentences from unrelated documents are treated as random negatives. Each training instance is represented as a quadruple (q, D, s, y). To obtain balanced and diverse supervision, we mix positive examples with both hard negatives and random negatives. In our implementation, this yields up to $150{,}000$ training examples and $7{,}000$ validation examples sampled from roughly 90k questions across the two datasets.

Why Teacher Distillation Helps. Hard labels alone provide limited supervision, since many non-supporting sentences may still appear superficially relevant to the question. In practice, training only with binary labels can encourage the model to over-rely on shallow cues such as entity overlap. Teacher-provided soft relevance scores offer graded supervision that helps the student distinguish clearly irrelevant sentences from harder negatives that are topically related but not answer-bearing.

Model Architecture and Training Setup. The student DistillSRE model is implemented as a transformer-based sequence classifier. Each input contains the question, the retrieved context surrounding the candidate sentence, and the candidate sentence itself; the model outputs a binary relevance score indicating whether the sentence is useful for answering the question in context. To adapt the model efficiently, we use parameter-efficient fine-tuning with LoRA adapters. Only the low-rank adaptation parameters and the classification head are updated, while the base model weights remain frozen. We also apply class-balanced weighting during optimization to mitigate label imbalance between positive and negative examples.

Inference Budget and Context Construction. At inference time, DistillSRE is applied to candidate sentences extracted from passages reranked by FUSE-QA. To construct the final evidence context for answer generation, we retain the highest-scoring sentences subject to a fixed budget. In our implementation, we keep up to $M = 3$ sentences with a maximum combined length of approximately 500 characters. This compact evidence set is typically sufficient for both POPQA and ENTITYQUESTIONS, where answers are often supported by one or a few sentences. The selected sentences are concatenated and passed to the generator as the final evidence context.

B Additional Experimental Details

This appendix provides implementation and evaluation details omitted from the main paper for space reasons. We first describe the benchmark datasets and entity-frequency tier construction, then present the generation prompts and evaluation protocol, followed by reproducibility details for dense index construction, and finally provide a worked end-to-end example of the full pipeline.

B.1 Dataset Details and Tier Construction

We evaluate on two open-domain QA benchmarks grounded in Wikipedia: POPQA and ENTITYQUESTIONS. Table 4 summarizes their basic statistics.

To evaluate long-tail behavior, we group questions according to the popularity of their subject entity, following the same tier definition used in RPDR and POPQA. We use three entity-frequency tiers: **Long-tail** ($s_{pop} < 100$), **Infrequent** ($100 \leq s_{pop} < 10{,}000$), and **Frequent** ($s_{pop} \geq 10{,}000$).

PopQA. POPQA provides a subject popularity value (s_pop) for each question. We directly assign each question to one of the three tiers using the thresholds above.

EntityQuestions. ENTITYQUESTIONS does not provide subject popularity annotations directly. To construct comparable tiers, we estimate subject popularity using Wikipedia pageview statistics. For each subject entity, we aggregate pageviews over a one-year period and use the resulting count as the popularity score s_{pop}. We then apply the same thresholds as above. The resulting tier distribution is reported in Table 2 of the main paper.

Property	PopQA	EntityQuestions
Knowledge Corpus	Wikipedia	Wikipedia
Dataset Statistics		
# Relations	16	24
Dataset Size (# Questions)	≈14K	≈220K
Training Set	N/A	176.6K
Development Set	N/A	22.1K
Test Set	14.3K	22.1K

Table 4: Basic statistics of the datasets.

Tier	s_{pop}	Example question
Long-tail	< 100	Who directed *Silent Sun*?
Infrequent	100–10,000	Where is Lake Prespa?
Frequent	$\geq 10{,}000$	Who wrote *Romeo and Juliet*?

Table 5: Example questions from each entity-frequency tier.

B.2 Generation Prompting and Evaluation Protocol

To ensure fair comparison across generators, all models are evaluated using the same answer-generation prompt template within each evaluation setting. All retrieval-based runs use the same evidence insertion format, and all retrieval-free runs use the same question-only format. For **FUSE-QA + DistillSRE**, the same sentence budget is used uniformly across all reported experiments.

Retrieval-based prompting. When external evidence is provided, we use the following instruction-style prompt:

Retrieval-based prompt

```
You are a factual QA assistant.
Use ONLY the provided context.
Return ONLY the answer as a short phrase (1–5 words).
If the answer is not explicitly stated in the context,
return UNKNOWN.
Make sure you pick the correct context.
Do not return the subject name when answering the
question.
Extract the answer from the given context.
Return the most likely correct answer.
No explanations. No full sentences.

Context:
{context}

Question: {question}
Answer:
```

Retrieval-free prompting. For retrieval-free evaluation, we use:

Retrieval-free prompt

```
You are a factual QA assistant.
Answer using only what you know.
Return ONLY the answer as a short phrase (1–5 words).
If you do not know, return UNKNOWN.
No explanations. No full sentences.

Question: {question}
Answer:
```

Context construction. For retrieval-based generation, the final context is built by concatenating the selected evidence snippets separated by blank lines. In the **FUSE-QA** setting, the context is formed directly from the reranked retrieved passages. In the **FUSE-QA + DistillSRE** setting, the context is formed from the top-ranked sentences selected by DistillSRE.

Uniform evaluation settings. All retrieval-based methods are evaluated under the same evidence budget. In the main experiments, DistillSRE retains at most $M = 3$ sentences with a total budget of approximately 500 characters. Shaded RPDR results in the main tables are copied directly from Zhang et al. (2025) rather than rerun in our codebase, and are included for reference under the original evaluation protocol.

B.3 Dense Index Construction

We build the dense index offline from English Wikipedia. For each entry, we concatenate the title and body text, truncate the input to the first 100 tokens, and encode it with the pretrained DPR context encoder `facebook/dpr-ctx_encoder-single-nq-base`. We use the pooled output as the dense representation, convert embeddings to `float32`, apply ℓ_2 normalization, and index them with FAISS `IndexFlatIP`, making retrieval equivalent to cosine similarity after normalization. We also store compressed metadata containing document ids, titles, and raw texts.

B.4 Worked End-to-End Example

We illustrate the full pipeline with the question *In what city was John XIX born?* The gold answer is **Rome**. In this example, FUSE-QA retrieves both relevant and distracting passages, while DistillSRE correctly isolates the answer-bearing sentence. Table 6 shows the highest-scoring candidate sentences ranked by DistillSRE.

Question. *In what city was John XIX born?*

Gold answer. **Rome**

Algorithm 1 Dense index construction

Require: Wikipedia entries $\{(d_i, t_i, x_i)\}_{i=1}^{N}$, tokenizer τ, DPR encoder f_{ctx}, max length $L = 100$
Ensure: FAISS index $\mathcal{I}$ and metadata $\mathcal{M}$
Initialize embedding list $\mathcal{E} \leftarrow [\,]$
for each batch $\mathcal{B}$ of Wikipedia entries **do**
  Compose input strings: $u_i \leftarrow t_i \,\|\, \backslash\mathrm{n} \,\|\, x_i$ for all $(d_i, t_i, x_i) \in \mathcal{B}$
  Tokenize: $\mathbf{z}_i \leftarrow \tau(u_i; L)$
  Encode: $\mathbf{h}_i \leftarrow f_{\mathrm{ctx}}(\mathbf{z}_i)$
  Append: $\mathcal{E} \leftarrow \mathcal{E} \cup \{\mathbf{h}_i\}$
end for
Stack embeddings into a matrix: $\mathbf{H} \leftarrow \mathrm{Stack}(\mathcal{E})$
Normalize rows: $\mathbf{H} \leftarrow \mathrm{Norm}_{\ell_2}(\mathbf{H})$
Build FAISS index: $\mathcal{I} \leftarrow \mathrm{IndexFlatIP}(\mathbf{H})$
Store metadata: $\mathcal{M} \leftarrow \{(d_i, t_i, x_i)\}_{i=1}^{N}$
Save $\mathcal{I}$ and $\mathcal{M}$ to disk

Rank	Candidate sentence	Score
1	Born Romanus in **Rome** was the third son of Gregory I, Count of Tusculum and his wife Mary.	0.9546
2	Pope John XIX was Pope from May 1024 to his death in 1032.	0.3124
3	Before becoming a Pope, John XIX was the Metropolitan of Al Beheira in Egypt.	0.1087
4	He joined the Paromeos Monastery in the Nitrian Desert as a monk.	0.0413
5	Upon the death of Benedict, Romanus was elected to succeed him.	0.0298

Table 6: Top candidate sentences ranked by DistillSRE for the worked example.

Selected evidence. *Born Romanus in **Rome** was the third son of Gregory I, Count of Tusculum and his wife Mary.*

Final prediction. **Rome**

C AI Assistants In Writing

We utilized ChatGPT to identify writing issues and polish the manuscript for clarity and conciseness.